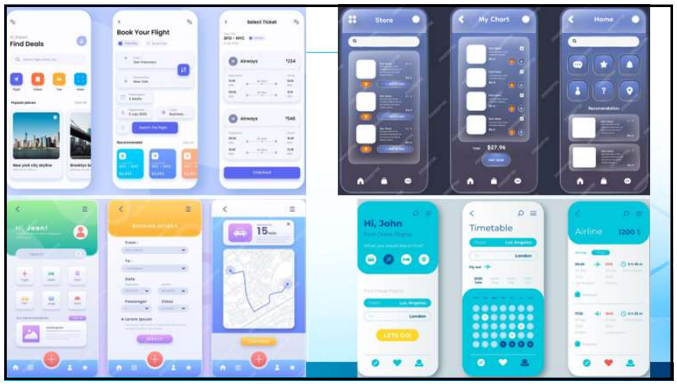
COS30017. SOFTWARE DEVELOPMENT FOR MOBILE DEVICES

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Email: kdnguyen@swin.edu.au
Phone: 0974.844.618

1



2

Week 1

- Canvas
- Syllabus
- Assignments and due date

3

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	14 Sep	Overview Designing for mobile	Review previous experience with OOP and XML, set up Android Studio, and GitHub
2	21 Sep	Kotlin	Portfolio components are due as set out in Canvas. • Test 1 in Week 3 labs. • Demonstrations for A1 in Weeks 4/5, due Week 4. **Hard deadlines will apply.**
3	28 Sep	App Building Blocks	
4	5 Oct	Lifecycles & States	
5	12 Oct	UX for Mobile	
6	19 Oct	Activity Communication and Fragments	
7	26 Oct	Lists	Portfolio components are due as set out in Canvas. Hard deadlines will apply.
8	2 Nov	Concurrent Programming	
9	9 Nov	Persistent Data	
10	16 Nov	Extra Topic	• Demonstrations for A2 in Weeks 7-9, due Week 8. • Test 2 in Week 10 labs (may change). • Progress for A3 in Weeks 11-12, due Week 12. **Hard deadlines will apply.**
11	23 Nov	Summary and Further Work	
12	30 Nov	Unit Retrospective	

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Assessment Due Date
1. Portfolio Assignments (A1 and A2)	Individual	40%	1,2,3	A1: End of Week 4 A2: End of Week 8
2. Portfolio Test 1	Individual	5%	1,2,3	End of Week 3
3. Portfolio Test 2	Individual	15%	1,2,3	End of Week 10
4. Project (A3)	Individual	40%	1,2,3	Progress Report: End of Week 11 Final Report & Investigation Video: End of Week 12

4

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	05 Jan	Overview Designing for mobile	Review previous experience with OOP and XML, set up Android Studio, and GitHub
2	12 Jan	Kotlin	Portfolio components are due as set out in Canvas. • Test 1 in Week 3 labs. • Demonstrations for A1 in Weeks 4/5, due Week 4. **Hard deadlines will apply.**
3	19 Jan	App Building Blocks	
4	26 Jan	Lifecycles & States	
5	02 Feb	UX for Mobile	
Lunar New Year Holiday from 08-Feb to 22-Feb inclusively			
6	23 Feb	Activity Communication and Fragments	
7	02 Mar	Lists	Portfolio components are due as set out in Canvas. Hard deadlines will apply.
8	09 Mar	Concurrent Programming	
9	16 Mar	Persistent Data	
10	23 Mar	Extra Topic	• Demonstrations for A2 in Weeks 7-9, due Week 8. • Test 2 in Week 10 labs (may change). • Progress for A3 in Weeks 11-12, due Week 12. **Hard deadlines will apply.**
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12	06 Apr	Unit Retrospective	

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This semester

5

Assessment

This is current for 2022 and will change in 2023. The portfolio contents will be clarified at the start of semester.

Assessment

This is a portfolio unit, so there is no final exam.

You are expected to complete tasks throughout semester, and submit them for feedback/completion, rather than leaving everything to the end of semester.

Note that submitted assessment(s) determined to be not of original work will be marked as "Not meeting expectations".

Core and extension tasks

These consists of basic assignments, as well as intermediate assignments for selected topics.

Core tasks involve some programming and short written tasks, while extension tasks involve more complex code extending from the core tasks and a technical report.

Note discussion posts and quizzes will also contribute towards competency.

Custom app level 1

For level 1, you are required to produce a fully worked through design for a mobile app, in addition to the core and extension tasks.

I recommend discussing proposed custom app with the convenor, the earlier the better. Note task submission and progress reports must be up to date before any discussion or agreement will take place. Progress reports are required during weeks 7-11.

Students who fall behind on core tasks will be asked to provide a learning plan for further support.

Custom app level 2/3

For a high distinction, you are required to produce a non-trivial app in addition to the basic and intermediate tasks. It is assumed that your project includes a fully worked through design in line with the level 1 requirements.

I recommend discussing the proposed app with your lecturer, the earlier the better. Progress reports are required during weeks 7-11.

A progress presentation needs to be made in week 12, ahead of the interview in week 14.

Final submission

You must submit a portfolio to pass this unit. This is a document with:

- Learning Summary Report (~6 pages)
- (links to) Portfolio items (these are made up of core assignment tasks, short reports, experience reports and videos).

Students aiming for a D or HD grade will be asked to attend an interview in week 13.

6

The valid use of genAI in this unit is as follows:

- **Output** of generative AI tools is **not** permitted to be submitted for assessment. Generative AI can be utilised for brainstorming and generating ideas to enhance work productivity. This includes the use of Gemini in the Android Studio IDE.
- Any use of genAI must be acknowledged, with prompts and outputs included in an appendix. If genAI was not used, the statement "No Generative AI tools were used for this task" must be included in the acknowledgement.

7

Unit resources

- Bryan Sills, Brian Gardner, Kristin Marsicano, Chris Stewart, Android Programming: The Big Nerd Ranch Guide, 5th Edition.
- John Horton, Android Programming with Kotlin for Beginners
- Peter Sommerhoff, Kotlin for Android App Development
- Wallace Jackson, Android Apps for Absolute Beginners
- Nate Ebel, Mastering Kotlin
- Svetlana Isakova, Dmitry Jemerov, Kotlin in Action
- Cay S.Horstmann, Core Java, Volume I Fundamentals, 11th Edition

8

Spike reports



[Your Name]
[DMS/JIRA]

Task ##P/C – Spike: [Insert Title]

Goals:

This section is an overview highlighting what the task is aiming to teach or upskill

Insert summary here
eg. Database handling and...

The following list outlines the goal broken down into more specific knowledge gaps involved in the goal

- Insert list here
- E.g. Database Indexing
- E.g. Database Storage
- ...

Tools and Resources Used

This section lists related software, tools, libraries, API's, and other resources used for this knowledge gap

- e.g. XCode
- e.g. (Link to Online Article - do not include things like "Google")
- ...

Knowledge Gaps and Solutions

This section presents the listed knowledge gaps and their solutions with supporting images, screenshots and captions where appropriate/required.

Gap 1: ...

((Insert steps required to address his knowledge gap))

9

Week 1

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10

Week 1

Custom app level 2/3

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11

Community



12

Week 1

- Install Android Studio -- instructions for this can be found in module 1;
- Set up GitHub -- instructions for this can be found in module 1;
- Look at resources that might be useful;
- Brush up on your OOP knowledge; and
- Brush up on your XML knowledge.

13

Object-Oriented Programming (OOP)

https://www.w3schools.com/java/java_classes.asp

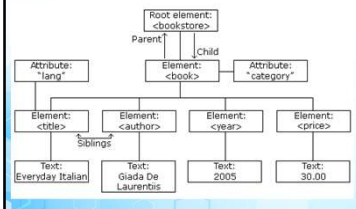
C3-OOP.pptx

C3-OOD.pptx

14

eXtensible Markup Language (XML)

https://www.w3schools.com/xml/




```
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  </book>
  <book category="children">
    <title lang="en">Harry Potter</title>
    <author>J K. Rowling</author>
    <year>2005</year>
    <price>29.99</price>
  </book>
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    <author>Erik T. Ray</author>
    <year>2003</year>
    <price>39.95</price>
  </book>
</bookstore>
```

XML Does Not DO Anything

15

Week 1

- First look at Android Studio
  - Android Studio Flamingo 2022.2.1



- The development tools we will use are:
  - Android SDK (Compiler and Emulator) -- the SDK compiles the code, the emulator runs as a separate program
  - Android Studio IDE -- this is where we write code and see where our files are
  - Gradle (Build System) -- this builds our apps
  - Debugger -- this allows us to step through our code
  - Visual UI Builder -- this allows us to drag and drop UI elements to create interfaces.

16

Set up GitHub -- The general git principles include:

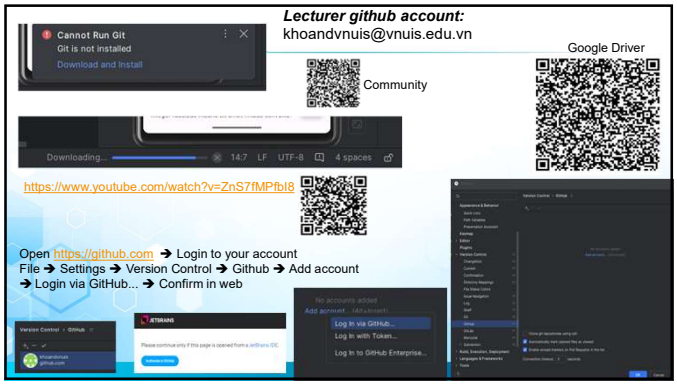
- a **repository** hosted online, which is the source for your project;
- you then **clone** the project onto your computer, either your personal machine or a lab machine;
- once this is done, you can work on the project, before **committing** your files with a comment saying what you have changed;
- the changes can then be **pushed** to the project, which means the repository online will be updated;
- if you are working from multiple locations, or with multiple people (for this unit, this could include tutors making edits to your code), you will need to **pull** the changes to your machine before continuing to work on the project.

17

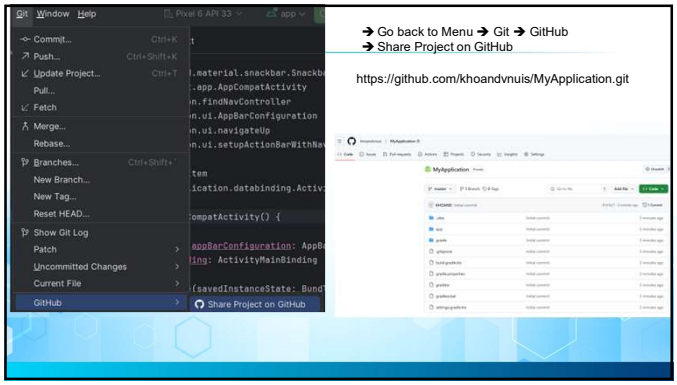
Week 1

- Set up GitHub -- To get started with an empty repo that you have a link for, and an existing project on your machine:
  1. Open the project in Android Studio.
  2. Under the **VCS** menu, select "***Enable version Control Integration***". Select **Git** from the drop down and click OK.
  3. You should now see a ***Git submenu under VCS***.
  4. Select Git -> Manage Remotes. Enter the URL of your Classroom repo.
  5. At some stage you'll be asked to enter your username and password so that a token can be generated for you (see instructions at the bottom of this page).
  6. Select Git -> Commit. Select all the files in app and write a message, then click Commit.
  7. You might get a warning if there are any problems with your code -- depending on the warnings these can be ignored.
  8. Select Git -> Push. Click Push in the bottom right corner. **This is very important; commit does not update your repository on GitHub.**

18



19



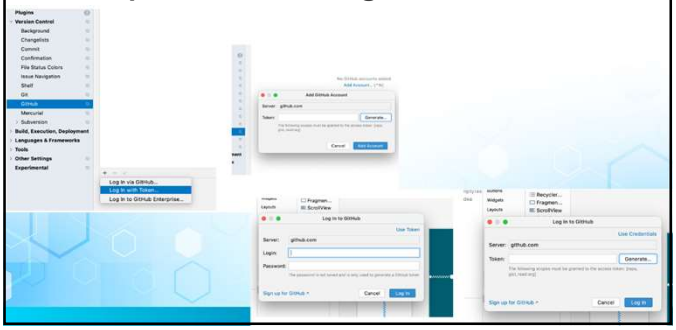
20

Week 1

- Set up GitHub -- In cases where a repo with existing code is provided to you (this could be a sample project):
  1. you can open this by going to VCS -> Get from Version Control -> select Git in the dropdown and entering the URL. You'll also need to specify where it will be stored on your machine.
  2. OR you can open this by going to File -> New -> Project from Version Control -> select Git in the dropdown and entering the URL. You'll also need to specify where it will be stored on your machine.
  3. OR you can select "Get project from Version Control" -> Git from the welcome screen. You'll need to enter the URL and also specify where it will be stored on your machine. This might be hiding in a drop down at the top right of the welcome screen.

21

Set up GitHub -- Getting a token

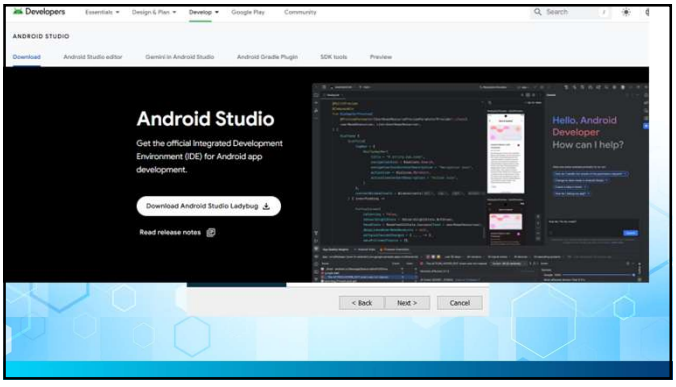


22

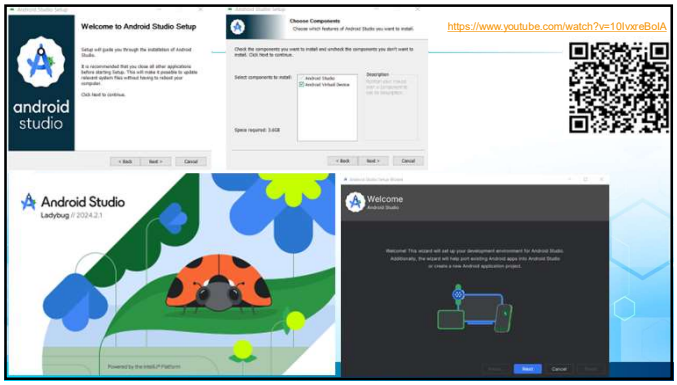
Set up GitHub -- Android Studio and GitHub testing

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  - Android Studio Flamingo 2022.2.1 (Android Studio Ladybug)
- The development tools we will use are:
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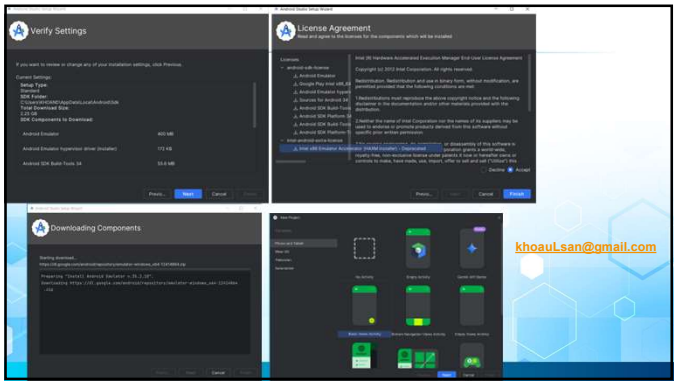
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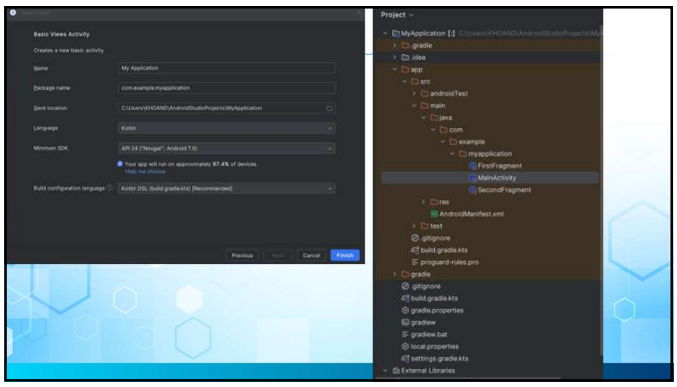
24


Welcome to Android Studio Setup
android studio
https://www.youtube.com/watch?v=10IvxreBolA
Android Studio
Ladybug
Welcome

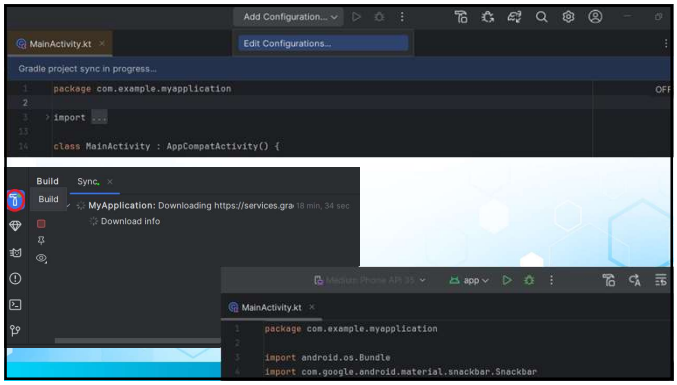
25


Verify Settings
License Agreement
Downloading Components
khoauLsan@gmail.com

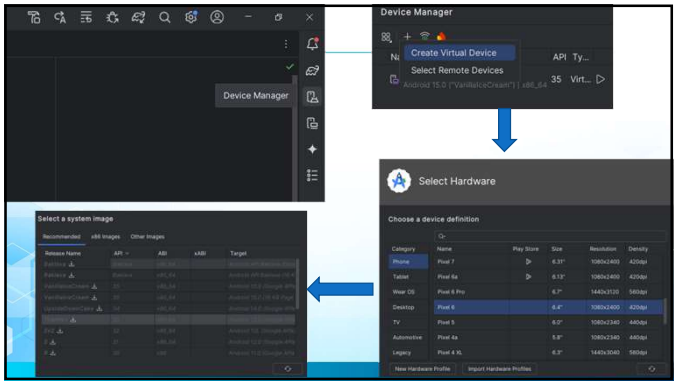
26


Basic Views Activity
Project

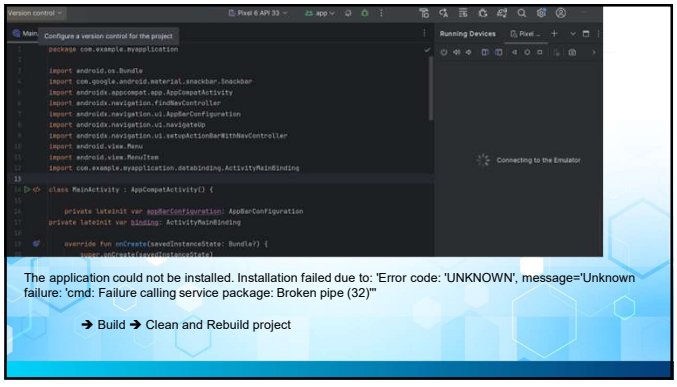
27


Add Configuration...
MainActivity.kt
Edit Configurations...
Gradle project sync in progress...
package com.example.myapplication
import ...
class MainActivity : AppCompatActivity() {
Build
Sync
Build
MyApplication: Downloading https://services.gra 18 min, 34 sec
Download info
app
MainActivity.kt
package com.example.myapplication
import android.os.Bundle
import com.google.android.material.snackbar.Snackbar

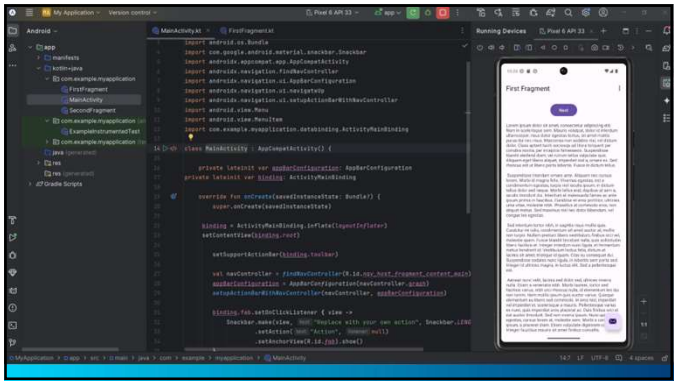
28


Device Manager
Create Virtual Device
Select Remote Devices
Device Manager
Select Hardware
Choose a device definition
Select a system image

29


Configure a version control for the project
Running Devices
Connecting to the Emulator
The application could not be installed. Installation failed due to: 'Error code: 'UNKNOWN', message='Unknown failure: 'cmd: Failure calling service package: Broken pipe (32)'"
➔ Build ➔ Clean and Rebuild project

30



31